

10(a)	(as temperature rises) electrons in valence band gain energy	B1
	electrons jump to conduction band	B1
	holes are left in the valence band	B1
	increased <u>number</u> (density) of charge carriers causes lower resistance	B1
10(b)(i)	$V^- = V^+$	C1
	$1.50 / 1.20 = R_T / 1.76$	C1
	$R_T = 2.2 \text{ (k}\Omega\text{)}$	C1
	temperature = 14 °C	A1
10(b)(ii)	(For LED to conduct,) V_{OUT} must be negative	B1
	$V^- > V^+$	B1
	R_T must be lower so temperature must be above (b)(i) value	B1